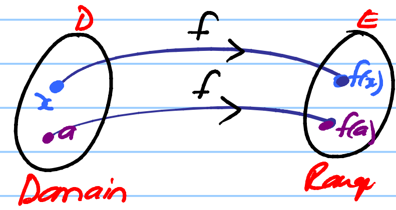
Note Title 2/8/2018

Unit 1.3. What is a function? (p.8-21)

Definition: (p.8)

A function f is a rule that assigns to each element x in a set D exactly one element $f(x)$ in a set E.

D: Domain of f
E: Range of f.



Eg: Area of circle with radius r:
$A(r) = \pi r^2$

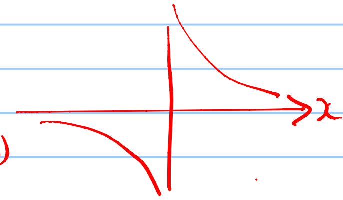
4 ways to present a function:

1. verbally (in words)
2. numerically (table)
3. visually (graph)
4. algebraically

See p. 10

Examples. Find the domain and range of

① $f(x) = \frac{1}{x}$

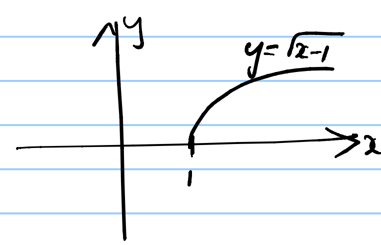


Domain: $D_f = (-\infty, 0) \cup (0, \infty)$
Range: $R_f = (-\infty, 0) \cup (0, \infty)$.

② $f(x) = \sqrt{x-1}$.

Domain: $x - 1 \geq 0$
$\Rightarrow x \geq 1$
$\Rightarrow x \in [1, \infty)$

Range: $R_f = [0, \infty)$



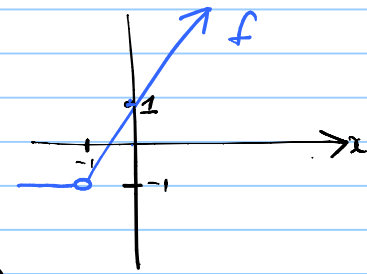
(3) Example 6 (p.13).

Piece-wise defined functions.

1) Examples 7, 8 (p.14-15)

2) $f(x) = x + |x+1|$

Find D_f, R_f and sketch f.



$$f(x) = x + |x+1|$$
$$= \begin{cases} x + x + 1 & \text{if } x+1 \geq 0 \\ x - (x+1) & \text{if } x+1 < 0 \end{cases}$$
$$= \begin{cases} 2x+1 & \text{if } x \geq -1 \\ -1 & \text{if } x < -1 \end{cases}$$

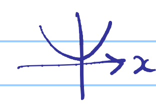
We use the vertical line test to determine if f is a function, given the graph of f. p.13

Symmetry of functions: $\boxed{\text{Consider } f(-x)}$ (p.16)

Even function: Symmetric about the Y-axis.

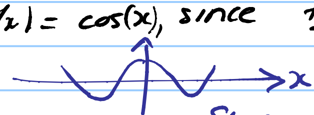
$f(-x) = f(x)$ for all $x \in D$.

Examples:



1) $f(x) = x^2$, since
$f(-x) = (-x)^2 = x^2 = f(x)$

2) $f(x) = \cos(x)$, since $f(-x) = \cos(-x) = \cos(x) = f(x)$

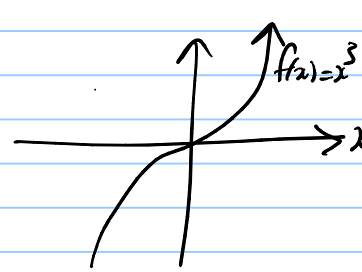


Symmetric about Y-axis

Odd function: Symmetric about the origin.

$\Rightarrow f(-x) = -f(x)$
for all $x \in D$

Examples: Consider



① $f(x) = x^3$

$f(-x) = (-x)^3 = -x^3 = -f(x)$
$\Rightarrow f$ is odd.

② $f(x) = \sin(x)$
$f(-x) = \sin(-x) = -\sin(x) = -f(x)$
$\Rightarrow f$ is odd.

Symmetric about the origin
(See p.16)

More examples: f odd, even or neither?

(a) $f(x) = 2x - x^2$

Always consider $f(-x)$!!

$$f(-x) = 2(-x) - (-x)^2$$
$$= -2x - x^2 \neq \begin{cases} f(x) \\ -f(x) \end{cases}$$

$\Rightarrow f$ is neither odd nor even.

(b) $f(x) = x|x|$

Always consider $f(-x)$ to determine if a function is odd/even.

$$f(-x) = -x\,|-x|$$
$$= -x|x|$$
$$= -f(x) \Rightarrow f \text{ is an } \underline{\text{odd}} \text{ function.}$$

Increasing/decreasing functions (p. 16)

- f is increasing on I if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$ on I
- f is decreasing on interval I if $f(x_1) > f(x_2)$ whenever $x_1 > x_2$ on I.